

Leonardo Perez

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PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate with hands-on experience in precision manufacturing, additive manufacturing process control, and simulation-driven design for research and competition hardware. Demonstrated ability to manufacture and inspect flight-critical and competition-winning components to tight dimensional tolerances. Experienced with LPBF systems, CNC machining, Wire EDM, ANSYS Fluent, and Siemens NX in both research and industry environments. 1st Place, America Makes Additive Manufacturing in Steel Competition; 2nd Place, International Rocket Engineering Competition (IREC).

EDUCATION

The University of Texas at El Paso

Bachelor of Science in Mechanical Engineering. Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026. GPA: 3.92/4.00.

Anticipated: May 2028

EXPERIENCE

Undergraduate Research Assistant

June 2025 – Present

W.M. Keck Center for 3D Innovation

El Paso, Texas

- Execute machine setup, powder sieving, and build preparation across multiple LPBF systems (EOS, SLM, Aconity, Renishaw, VELO3D) to ensure process-controlled conditions for each build, accumulating hands-on experience across 100+ processed samples.
- Machine components and custom fixtures for post-processing using a HAAS VF-2 CNC mill, programming toolpaths in Fusion 360 and SolidWorks to support internal and external LPBF research projects.
- Perform post-processing with Wire EDM (SSG HB400) to remove parts from build plates according to customer and research specifications, consistently meeting tolerances of ± 0.005 in.
- Operate, set up, and program a custom gantry system integrated into an RPMI 222XR using a Siemens Sinumerik controller, supporting Directed Energy Deposition (DED) research in a hybrid additive manufacturing environment.
- Develop and optimize CNC toolpaths in Siemens NX for hybrid additive manufacturing processes, ensuring accurate gantry motion and reliable deposition during experimental builds.
- Section DED specimens using Wire EDM (SSG HB400) to support metallurgical analysis, process validation, and characterization of hybrid manufactured components.

Engineering Intern

June 2026 – August 2026

Vinton Steel, LLC

El Paso, Texas

- Designed and optimized a cooling pipe for a hot rolling mill stand by developing and evaluating multiple concepts through ANSYS Fluent simulations, reducing overspray affecting adjacent equipment and extending bearing service life.
- Authored a comprehensive Standard Operating Procedure (SOP) for roll stand assembly, standardizing maintenance practices and improving consistency, efficiency, and training for Roll Shop personnel.
- Installed and programmed NFC tags on rolling mill rolls, encoding identification and inventory data to improve roll traceability and support a more efficient inventory management system.

Manufacturing Team Lead

January 2026 – Present

Sun City Summit Rocket Team

El Paso, Texas

- Lead a manufacturing team producing flight-critical hardware for UTEP's participation at the International Rocket Engineering Competition (IREC), coordinating machining operations and task execution.
- Independently manufacture flight-critical components including rocket fins, fin mounting brackets, a steel nose tip, ballast, and bulk plates using Wire EDM and CNC milling, consistently achieving dimensional tolerances of ± 0.01 in.
- Perform dimensional inspection and quality verification of manufactured components to ensure precise fit, alignment, and aerodynamic performance prior to final integration.

AWARDS & HONORS

- 1st Place, America Makes Additive Manufacturing in Steel Competition (2026): Selected as one of two undergraduates on the multidisciplinary team that designed and manufactured an LPBF stainless steel multi-tool; performed machine setup, build preparation, build removal, and Wire EDM post-processing for the competition-winning component.
- 2nd Place, International Rocket Engineering Competition (IREC), 30k COTS Category (2026): Selected as one of 14 representatives from a 100+ member team based on technical contributions; supported final assembly, integration, and launch operations at the competition.

SKILLS

Additive Manufacturing: LPBF systems (EOS, SLM 280, Renishaw, Aconity, VELO3D), powder handling and sieving, build setup, post-processing, Directed Energy Deposition (DED)

Subtractive Manufacturing: CNC machining (milling and turning), Wire EDM, manual mill and lathe, bandsaw

Software: Siemens NX, ANSYS Fluent, Fusion 360, SolidWorks (CAD/CAM), MATLAB, Microsoft Office

Languages: Fluent in English and Spanish